

Technology News

Robot the size of a molecule

In a significant advancement in the nascent fields of molecular computing and robotics, researchers in the US have created robots the size of a single molecule that can move independently across a nano-scale track. The achievement is expected to lead to molecular robots that can fix individual cells or assemble nanotechnology products.

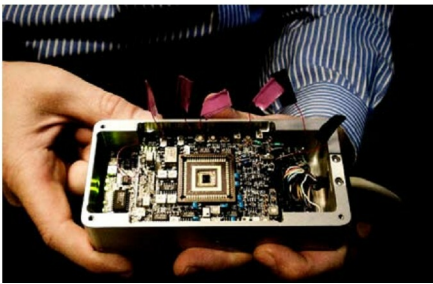
The project involved reprogramming DNA molecules to perform in specific ways. The strings of reprogrammed DNA so developed could move autonomously through a specially-created, two-dimensional landscape. These DNA walkers acted in rudimentary robotic ways, showing that they were capable of starting motion, walking for a while, turning and stopping.

These robots are a combined work of researchers from Columbia University, Arizona State University, the University of Michigan and the California Institute of Technology (Caltech).

Smart dust to monitor Earth

In the 1990s, a researcher named Kris Pister had thought of a future in which people would sprinkle the Earth with billions of tiny sensors, no larger than grains of rice. These 'smart dust' particles, as he called them, would then act as a camera, keeping an electronic watch over everything. Fitted with computers and sensors, they would relay real-time data about people, cities and the natural environment. Now, it seems that this fantasy is starting to become a reality.

Hewlett-Packard (HP) is working on a project called the Central Nervous System for the Earth. As part of this project, within the next few years, it plans to deploy a trillion sensors all over the planet. These devices would study the health of ecosystems, detect earthquakes more rapidly, predict



HP Labs researcher holds a prototype of a sensor network node incorporating the new MEMS accelerometer from HP. It's first to be deployed as part of HP Labs' Central Nervous System for Earth (CeNSE)

traffic patterns and monitor energy usage. The basic idea is that accidents could be prevented and energy saved if people knew more about the world in real time.

The firm will tie up with Royal Dutch Shell to install one million matchbook-size monitors to help in the oil exploration business by measuring rock vibrations and movement.

Wearable impact sensor for athletes

New Zealand researchers have developed a new wearable impact sensor that could soon provide much needed information about the stresses and strains on limbs for rugby players, high jumpers and runners and help them strive for that perfect performance. The sensing materials, known as ionic polymer metallic composites (IPMCs), produce an electrical current when compressed.

IPMCs are flexible, lightweight and durable and therefore can be fashioned into wearable sensor devices to monitor impact forces without interfering

with an athlete's performance. These are usually made from an ionic polymer, such as nafion or flemion, which is coated with a conducting metal, platinum or gold.

Applying a voltage to IPMC materials causes them to flex as ions migrate, causing electrostatic repulsion within the composite material. The opposite effect, in which ion movement generates a voltage when the material is flexed, is exploited in the sensor technology.

Impact sensors made from IPMC could be inserted into footwear to measure the impact energy of a foot striking a hard surface, or they might be placed in a rugby player's shoulder pads to measure collision impacts or forces exerted during a rugby scrum.

Car engines too can be hacked

Researchers from the University of California and University of Washington have demonstrated how computer systems embedded in cars to control their various functions can fall prey